

Color Pattern Formation in Butterfly Wings

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1 Introduction

If you have ever wondered how animals acquire their striking coat patterns, how seashells develop intricate striations, or how leaves are arranged along a stem, you may be surprised to learn that these phenomena may arise from a common underlying mechanism. In the early 1950s, Alan Turing introduced a mathematical framework for the spontaneous emergence of spatial patterns in biological systems. Although conceptually simple, his model was far ahead of its time and remained largely unexplored for decades. Turing proposed that spatial pattern formation can arise from interactions between two diffusible chemicals, or morphogens. Reaction–diffusion equations formalize this mechanism by describing how morphogen concentrations evolve through local reactions and spatial diffusion. Two canonical classes of such models emerged: activator–inhibitor systems and substrate–depletion systems. In an activator–inhibitor system, patterns form through the combination of local activation and long-range inhibition: nearby interactions amplify a signal, while more distant ones suppress it. Turing’s models are an intriguing way to capture how complex biological patterns might emerge, even if their full relevance remains to be confirmed. My investigation of color patterning on butterfly wings focuses on a particular activator–inhibitor model, the Gierer–Meinhardt system, building on the work of Sekimura et al. [2].

2 Canonical Mathematical Model

In an activator–inhibitor model, the activator promotes both its own production and that of the inhibitor, while the inhibitor suppresses itself and the activator. Let u and v denote the concentrations of the activator and inhibitor. Their dynamics are governed by the reaction–diffusion system

$$\begin{aligned}\frac{\partial u}{\partial t} &= f(u, v) + D_1 \nabla^2 u, \\ \frac{\partial v}{\partial t} &= g(u, v) + D_2 \nabla^2 v,\end{aligned}\tag{1}$$

where f and g describe the local reaction kinetics and the diffusion terms represent long-range chemical transport. Since u activates both itself and v , f and g should increase with u ; and since v inhibits both itself and u , they should decrease with v . We adopt the Gierer–Meinhardt kinetics, where $k_1 \dots k_7$ are positive rate constants:

$$\begin{aligned}f(u, v) &= k_1 - k_2 u + \frac{k_3 u^2}{v(k_6 + k_7 u^2)}, \\ g(u, v) &= k_4 u^2 - k_5 v.\end{aligned}\tag{2}$$

For simplicity, classical activator–inhibitor models assume a rectangular domain, homogeneous tissue response, and no-flux (Neumann) boundary conditions. Sekimura et al. relax these assumptions by imposing a spatially varying threshold, a polygonal approximation of the wing, and Robin boundary conditions. They numerically compare these patterns with those observed in *P. dardanus* [2]. However, they do not study the effects of each relaxed assumption in isolation or quantitatively analyze the extended model.

3 Proposed Extension

Sekimura et al. suggest that a scale cell's sensitivity to the patterning chemical may vary across the wing [2]. Motivated by this, I will investigate the isolated effect of a spatially varying threshold function, modeled as a plane

$$v \geq \alpha y + \beta x + c_0, \quad (3)$$

where $\alpha, \beta \neq 0$ and $c_0 \geq 0$. Biologically, this function represents positional variation in a cell's responsiveness to the inhibitor: if the local concentration v exceeds the threshold, cells become black; otherwise, they remain colored. Setting $\alpha = \beta = 0$ recovers the standard constant threshold. My investigation has two components. First, I will conduct a stability analysis to understand the usual interpretation of stability in Turing systems. Second, I will investigate how a spatially varying threshold function biases the visible patterns produced by the system and implement numerical simulations to explore those dynamics.

4 Stability Analysis

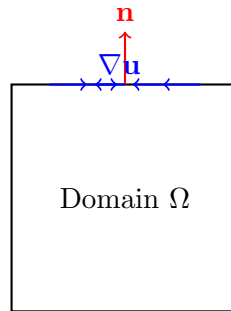
Turing showed that a homogeneous steady state (HSS) that is stable without diffusion can become unstable once diffusion is introduced, a phenomenon known as diffusion-driven instability. In the butterfly wing context, the HSS corresponds to a uniformly colored wing, while the unstable state corresponds to patterned coloration. To demonstrate this mechanism, I will carry out a linear stability analysis of the canonical reaction–diffusion model. Because my proposed extension affects only how the resulting chemical fields are interpreted as pigmentation on the butterfly-wing domain, and not the reaction–diffusion equations themselves, the usual stability interpretation suffices. This analysis is well known, and I will follow the notation used in Helen M. Byrne's lecture notes [1]. First, we will rewrite Equation 1 in matrix-vector notation. Let

$$\mathbf{u} = \begin{pmatrix} u \\ v \end{pmatrix}, \quad F(\mathbf{u}) = \begin{pmatrix} f(u, v) \\ g(u, v) \end{pmatrix}, \quad D = \begin{pmatrix} D_u & 0 \\ 0 & D_v \end{pmatrix}.$$

Then, we write the problem with homogeneous Neumann boundary conditions as

$$\mathbf{u}_t = D \nabla^2 \mathbf{u} + F(\mathbf{u}), \quad \mathbf{n} \cdot \nabla \mathbf{u} = 0, \quad x \in \partial\Omega. \quad (4)$$

The homogeneous Neumann boundary conditions specify zero flux across the boundary; that is, no chemicals enter or leave the butterfly-wing domain. The below diagram is a caricature of zero-flux boundary conditions on one edge of a square domain.



Let u_s be the homogeneous steady state (HSS), i.e., a spatially uniform constant solution satisfying $F(u_s) = 0$. We can understand the behavior of our nonlinear system near the HSS by linearizing about it. Let $u = u_s + w$, where w is a small perturbation (i.e., $\|w\| \ll 1$). First, we will Taylor expand $F(u)$ about the HSS.

$$\begin{aligned}
F(u) &= F(u_s + w) \\
&= F(u_s) + F'(u_s)w + O(\|w\|^2) \\
&= F'(u_s)w + O(\|w\|^2)
\end{aligned}$$

where $F'(u_s) = J$ is the Jacobian evaluated at the HSS. Next, we will substitute this Taylor expansion into our PDE, dropping higher-order terms.

$$u_t = D\nabla^2 u + F(u) \approx D\nabla^2 w + Jw.$$

Since the linearized equation is linear in w , we seek solutions of the separable form $w(x, t) = A(t)p(x)$, and by the principle of superposition, the most general solution is a linear combination of these separable solutions. Next, we will substitute this ansatz into our linearized PDE.

$$\begin{aligned}
A'(t)p(x) &= DA(t)\nabla^2 p(x) + JA(t)p(x), \\
\frac{A'(t)}{A(t)} &= \frac{D\nabla^2 p(x) + Jp(x)}{p(x)}.
\end{aligned}$$

Since the left-hand side depends only on t while the right-hand side depends only on x , and these are independent variables, both sides must equal the same constant, which we denote by λ .

$$\begin{aligned}
\frac{A'(t)}{A(t)} &= \lambda, \\
D\nabla^2 p(x) + Jp(x) &= \lambda p(x).
\end{aligned}$$

The first equation is simply a one-dimensional linear ODE, whose solution is an exponential. Hence, $A(t) = A_0 e^{\lambda t}$.

Ansatz: $\nabla^2 p = -k^2 p$. This converts the spatial PDE into an algebraic eigenvalue problem by choosing p to be an eigenfunction of the Laplacian. The allowed eigenfunctions and eigenvalues k^2 are determined by the domain geometry and boundary conditions. Then, we substitute the ansatz into the spatial equation.

$$\begin{aligned}
D(-k^2)p + Jp &= \lambda p, \\
(-Dk^2 + J)p &= \lambda p.
\end{aligned}$$

For p not identically equal to 0:

$$\det(\lambda I - J + Dk^2) = 0,$$

$$\lambda^2 - \lambda \text{tr}(J - Dk^2) + \det(J - Dk^2) = 0,$$

$$\lambda_{1,2}(k^2) = \frac{\text{tr}(J - Dk^2) \pm \sqrt{[\text{tr}(J - Dk^2)]^2 - 4 \det(J - Dk^2)}}{2}. \quad (5)$$

The general separable solution is

$$w(x, t) = A_0 e^{\lambda t} p(x).$$

By the principle of superposition, the most general solution is a linear combination of these separable solutions. Hence, the most general solution is

$$w(x, t) = \sum_k A_k e^{\lambda(k^2)t} p_k(x).$$

Here k^2 is the eigenvalue, or equivalently k is the wave number.

Showing Stability Without Diffusion

When diffusion is absent, $k^2 = 0$. We will substitute $k^2 = 0$ into Equation 5.

$$\lambda_{1,2}(0) = \frac{(f_u + g_v) \pm \sqrt{(f_u + g_v)^2 - 4(f_u g_v - f_v g_u)}}{2}.$$

For stability, we need $\text{Re}(\lambda_{1,2}) < 0$. Since the square root term is always non-negative, the sum $(f_u + g_v)$ must be negative to ensure both eigenvalues have negative real parts. This gives two conditions:

$$f_u + g_v < 0 \quad (1),$$

$$f_u g_v - f_v g_u > 0 \quad (2).$$

Condition (1) follows directly from requiring $\text{Re}(\lambda_{1,2}) < 0$. Condition (2) ensures the determinant of the Jacobian is positive. Since the determinant equals the product of the eigenvalues, a positive determinant means both eigenvalues have the same sign. Combined with condition (1), this guarantees both eigenvalues are either real and negative, or complex conjugates with negative real parts.

Showing Instability With Diffusion

$$\lambda_{1,2}(k^2) = \frac{\text{tr}(J - Dk^2) \pm \sqrt{[\text{tr}(J - Dk^2)]^2 - 4 \det(J - Dk^2)}}{2}.$$

We know by (1) and $D_u, D_v > 0$ that $f_u + g_v - (D_u + D_v)k^2 < 0$. We need $\text{Re}(\lambda_{1,2}(k^2)) > 0$. Because the trace is negative, in order for the numerator to be positive, we need:

$$\sqrt{[\text{tr}(J - Dk^2)]^2 - 4 \det(J - Dk^2)} > |f_u + g_v - (D_u + D_v)k^2|.$$

Squaring both sides:

$$(f_u + g_v - (D_u + D_v)k^2)^2 < [\text{tr}(J - Dk^2)]^2 - 4 \det(J - Dk^2). \quad (6)$$

Since the left-hand side equals $[\text{tr}(J - Dk^2)]^2$, this simplifies to:

$$[\text{tr}(J - Dk^2)]^2 < [\text{tr}(J - Dk^2)]^2 - 4 \det(J - Dk^2),$$

$$0 < -4 \det(J - Dk^2),$$

$$\det(J - Dk^2) < 0.$$

Let $h(k^2) = \det(J - Dk^2)$.

$$h(k^2) = \det(J - Dk^2) = D_u D_v k^4 - (D_v f_u + D_u g_v)k^2 + (f_u g_v - f_v g_u) < 0.$$

For instability, we need $h(k^2) < 0$ for some range of k^2 values. Note that $f_u g_v - f_v g_u > 0$ by condition (2), and the coefficient of k^4 is positive. Hence,

$$D_v f_u + D_u g_v > 0 \quad (3).$$

Since $h(k^2) < 0$, we have $h(k_{\min}^2) < 0$.

$$\frac{dh}{dk^2} = 2D_u D_v k^2 - (D_v f_u + D_u g_v) = 0,$$

$$k_{\min}^2 = \frac{D_v f_u + D_u g_v}{2D_u D_v}.$$

Substituting $k_{\min}^2$ into the condition $h(k_{\min}^2) < 0$:

$$h(k_{\min}^2) = D_u D_v \left(\frac{D_v f_u + D_u g_v}{2D_u D_v} \right)^2 - (D_v f_u + D_u g_v) \left(\frac{D_v f_u + D_u g_v}{2D_u D_v} \right) + f_u g_v - g_u f_v < 0,$$

$$\frac{(D_v f_u + D_u g_v)^2}{4D_u D_v} - \frac{(D_v f_u + D_u g_v)^2}{2D_u D_v} + f_u g_v - g_u f_v < 0,$$

$$-\frac{(D_v f_u + D_u g_v)^2}{4D_u D_v} + f_u g_v - g_u f_v < 0,$$

$$(f_u g_v - g_u f_v) 4D_u D_v < (D_v f_u + D_u g_v)^2,$$

$$2\sqrt{D_u D_v (f_u g_v - g_u f_v)} < D_v f_u + D_u g_v \quad (4).$$

Notes / Interpretations

1. To have both $D_v f_u + D_u g_v > 0$ (3) and $f_u + g_v < 0$ (1), we need $D_u \neq D_v$.
2. To have both $D_v f_u + D_u g_v > 0$ (3) and $f_u g_v - f_v g_u > 0$ (2), f_u and g_v must have opposite signs. In particular, for the activator-inhibitor model, we want:

$$J = \begin{pmatrix} + & - \\ + & - \end{pmatrix}.$$

From the Jacobian signs, $f_u > 0$ and $g_v < 0$. Condition (3) then requires $D_v > D_u$: the activator must diffuse more slowly than the inhibitor.

Nondimensionalization

The dimensional system is

$$\frac{\partial u}{\partial t} = k_1 - k_2 u + \frac{k_3 u^2}{v(k_6 + k_7 u^2)} + D_1 \nabla^2 u,$$

$$\frac{\partial v}{\partial t} = k_4 u^2 - k_5 v + D_2 \nabla^2 v.$$

Reference scales

$$t_{\text{ref}} = \frac{1}{k_5},$$

$$L_{\text{ref}} = \sqrt{\frac{D_1}{k_5}},$$

$$u_{\text{ref}} = \frac{k_3}{k_4 k_6},$$

$$v_{\text{ref}} = \frac{k_4}{k_5} u_{\text{ref}}^2.$$

Nondimensional variables $\hat{u}, \hat{v}, \hat{t}, \hat{x}$ give

$$\begin{aligned}\frac{\partial \hat{u}}{\partial \hat{t}} &= a - \frac{b}{\gamma} \hat{u} + \frac{\hat{u}^2}{\hat{v}(1 + K\hat{u}^2)} + \hat{\nabla}^2 \hat{u}, \\ \frac{\partial \hat{v}}{\partial \hat{t}} &= \hat{u}^2 - \hat{v} + d \hat{\nabla}^2 \hat{v},\end{aligned}$$

with

$$\gamma = \frac{k_5}{k_2}, \quad a = \frac{k_1}{k_5 u_{\text{ref}}}, \quad K = \frac{k_7 u_{\text{ref}}^2}{k_6}, \quad d = \frac{D_2}{D_1}, \quad b = 1.$$

Now, we let $\hat{t} = \gamma t$, $\hat{x} = \sqrt{\gamma} x$. Then, after rearrangement the equations become (drop hats)

$$\begin{aligned}\frac{\partial u}{\partial t} &= \nabla^2 u + \frac{1}{\gamma} \left(a - bu + \frac{u^2}{v(1 + Ku^2)} \right), \\ \frac{\partial v}{\partial t} &= d \nabla^2 v + \frac{1}{\gamma} (u^2 - v).\end{aligned}$$

Note that we can easily change $\frac{1}{\gamma}$ to γ , which simply changes the interpretation of the parameter from a small to a large multiplier of the reaction terms.

Computing the Homogeneous Steady State Using Newton's Method

We now numerically compute the homogeneous steady state (HSS) of the nondimensionalized Gierer-Meinhardt system. The steady state is found by solving the system of equations:

$$\begin{aligned}f(u, v) &= a - bu + \frac{u^2}{v(1 + Ku^2)} = 0, \\ g(u, v) &= u^2 - v = 0.\end{aligned}$$

Following Sekimura et al. [2], we fix the parameter values $a = 0.1$, $b = 1.0$, and $K = 0.5$. For these parameters, the steady state equations are nonlinear and must be solved numerically. We employ the multidimensional Newton-Raphson method, which iteratively solves:

$$J(\mathbf{x}^{(n)}) \Delta \mathbf{x}^{(n)} = -F(\mathbf{x}^{(n)})$$

where $\mathbf{x}^{(n)} = (u^{(n)}, v^{(n)})$ is the solution at iteration n , and the update is given by $\mathbf{x}^{(n+1)} = \mathbf{x}^{(n)} + \Delta \mathbf{x}^{(n)}$. Here,

$$F(\mathbf{x}) = \begin{pmatrix} f(u, v) \\ g(u, v) \end{pmatrix}$$

is the vector of functions whose root we seek, and

$$J(\mathbf{x}) = \begin{pmatrix} f_u & f_v \\ g_u & g_v \end{pmatrix}$$

is the Jacobian matrix evaluated at $\mathbf{x} = (u, v)$.

The partial derivatives are:

$$\begin{aligned}f_u &= -b + \frac{2u}{v(1 + Ku^2)} - \frac{2Ku^3}{v(1 + Ku^2)^2}, \\ f_v &= -\frac{u^2}{v^2(1 + Ku^2)}, \\ g_u &= 2u, \\ g_v &= -1.\end{aligned}$$

Starting from an initial guess $(u^{(0)}, v^{(0)})$, we iterate until $\|F(\mathbf{x}^{(n)})\| < \epsilon$ for some tolerance ϵ . For the given parameter values and a tolerance of $\epsilon = 10^{-10}$, my implementation of Newton's method converges in 12 iterations to the steady state $(u_s, v_s) = (0.839457, 0.704688)$. Sekimura et al. [2] report the steady state $(u_s, v_s) = (0.8395, 0.7047)$. The full implementation can be found in `steadystate.py`. All code in this project was initially generated with an LLM and subsequently refined manually.

Plotting Nullclines

Next, I wrote a program to visualize the nullclines and phase plane dynamics using `matplotlib`. Figure 1 shows the resulting plot, which displays both nullclines, the steady state, and the direction field. The full implementation can be found in `nullclines.py`.

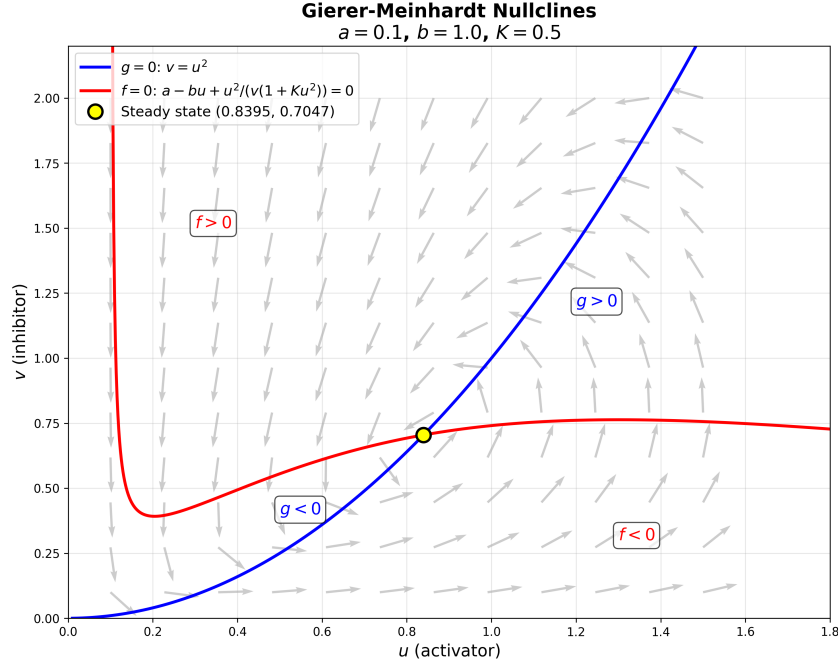


Figure 1: Phase Plane Portrait of Nullclines

Numerical Simulations of Butterfly Wing

Spatial modes for Neumann Boundary Conditions

Recall that in the linearized PDE, each separable solution evolves as

$$w(x, t) = A_0 e^{\lambda t} p(x),$$

with $\lambda = \lambda(k^2)$ given by Equation (5). The spatial structure of $p(x)$ is determined by the zero-flux boundary conditions on the rectangular domain. Under these conditions, the allowed eigenfunctions of the Laplacian are products of cosine functions, which we denote as $p_{m,n}(x, y)$ for integers $m, n \geq 0$. The general solution can then be written as a sum over these modes,

$$w(x, t) = \sum_{m,n} A_{m,n} e^{\lambda(k_{m,n}^2)t} p_{m,n}(x),$$

where $k_{m,n}^2$ is the corresponding eigenvalue. Each (m, n) mode evolves independently, with growth or decay determined by the linear stability criterion: if $\text{Re}(\lambda(k_{m,n}^2)) > 0$, the mode grows; if $\text{Re}(\lambda(k_{m,n}^2)) < 0$, it decays.

In numerical simulations, it is convenient to represent the solution using a discrete cosine transform (DCT). The DCT expresses the solution as a sum of cosine modes, allowing each mode to be tracked and updated separately, which makes the computations faster and simpler. This approach also makes it easy to focus on specific modes, such as the $(m, n) = (3, 0)$ mode, by choosing parameters (d, γ) to make that mode grow more strongly than the others.

Numerical Simulations

I carried out two numerical simulations to investigate color pattern formation on the butterfly wing using the Gierer-Meinhardt activator-inhibitor system. The first simulation employs a constant expression threshold,

$$v \geq 0.69,$$

while the second uses a spatially varying (planar) threshold of the form

$$v \geq -0.111 y - 0.025 x + 0.69.$$

In both cases, regions in which the inhibitor concentration v exceeds the chosen threshold are interpreted as black markings, whereas regions below the threshold correspond to the background color. The simulations were performed on a square domain with homogeneous Neumann (zero-flux) boundary conditions. To discretize the system, I used a semi-implicit spectral method based on the two-dimensional discrete cosine transform (DCT-II). At each timestep, the non-linear reaction terms are evaluated explicitly in physical space, while the linear diffusion terms are treated implicitly in Fourier space. This enables significantly larger timesteps than would be possible with a fully explicit scheme. A crucial modeling decision concerns the choice of the initial perturbation. Sekimura et al. report that for parameter values in a neighborhood of

$$(d, \gamma) \approx (70.85, 619.45),$$

the uniform steady state becomes linearly unstable primarily to spatial perturbations of the form $\cos(3\pi x)$, corresponding to the $(3, 0)$ mode [2]. To reflect this mode-selection mechanism, the simulations are initiated with a small-amplitude perturbation of the form

$$u_0(x, y) = u_s(1 + \varepsilon \cos(3\pi x)), \quad v_0(x, y) = v_s(1 + \varepsilon \cos(3\pi x)),$$

where (u_s, v_s) is the homogeneous steady state.

Constant Threshold Function

Figure 2 shows the results of the simulation with a constant threshold. The full code for this simulation can be found in `turingpatterns-constant.py`. In the first three snapshots, the entire “butterfly wing” is uniformly black, indicating that the inhibitor concentration $v(x, t)$ is everywhere above the fixed threshold. Although the underlying PDE dynamics are already evolving, the amplitude of the $(3, 0)$ cosine perturbation is initially too small to push any region of v below the threshold. As a result, the plot appears unchanged across these early times. By $t = 10$, however, we see four vertical bands: black, yellow, black, and yellow. The number of bands is determined by the $(3, 0)$ spatial mode, which completes one and a half cosine periods across the domain. The inhibitor concentration is represented by

$$v(x, t) = v_s + A(t) \cos(3\pi x),$$

where v_s denotes the inhibitor component of the homogeneous steady state (HSS). The peaks of the cosine add to v_s , while the troughs subtract from it. Once the amplitude $A(t)$ grows large enough, the troughs fall below the threshold, creating yellow bands. The apparent suddenness of the pattern, with the first three snapshots looking identical and the last three snapshots also looking identical, is a consequence of the binary nature of thresholding. The inhibitor field v evolves smoothly, but the visualization converts it into a two-color map, making the transition appear abrupt.

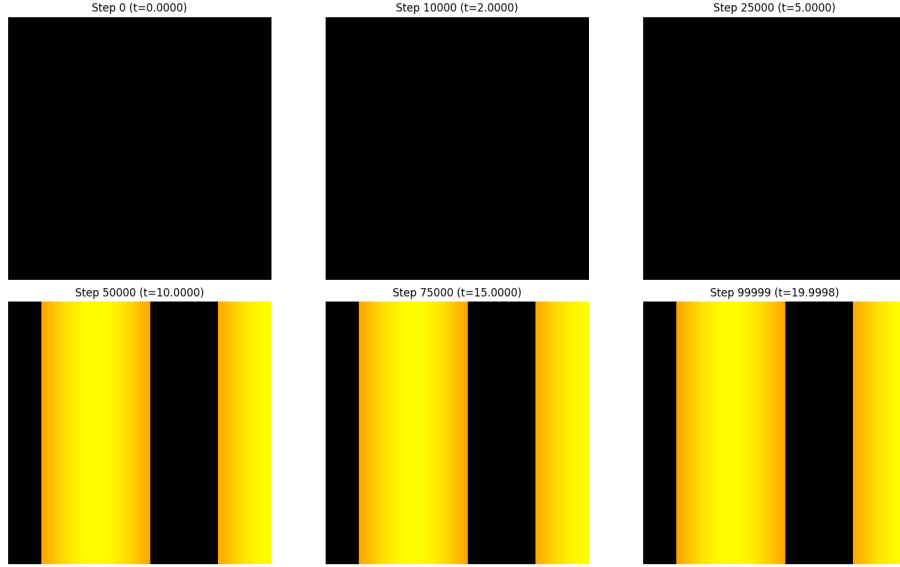


Figure 2: Time Series Plot of Turing Pattern Formation with Constant Threshold

Planar Threshold Function

Figure 3 shows the results of the simulation with a planar threshold. The full code for this simulation can be found in `turingpatterns-planar.py`. The main difference compared to the previous figure is that we no longer see four vertical bands. Instead, the alternating black and yellow regions now form a series of hills and valleys, with the yellow regions corresponding to the hills. This pattern arises from the nonlinear “decision boundary” imposed by the planar threshold in the binary color map. The planar threshold at each spatial location is given by

$$\text{threshold}(x, y) = c_0 + \alpha y + \beta x.$$

The decision boundary is defined as the curve where $v(x, y) = \text{threshold}(x, y)$. Solving for y , we obtain

$$y(x) = \frac{v(x) - c_0 - \beta x}{\alpha}.$$

Intuitively, this curve separates regions where the inhibitor concentration exceeds the threshold (black) from regions where it is below the threshold (yellow). Because $\alpha < 0$, lower concentrations in the troughs of $v(x)$ produce higher boundary values, creating the upward hills in the yellow regions, while higher concentrations in the peaks produce lower boundary values, resulting in the valleys in black. This analytical boundary has been plotted on the figure for reference. The boundary slightly deviates from the true plot due to numerical precision artifacts.

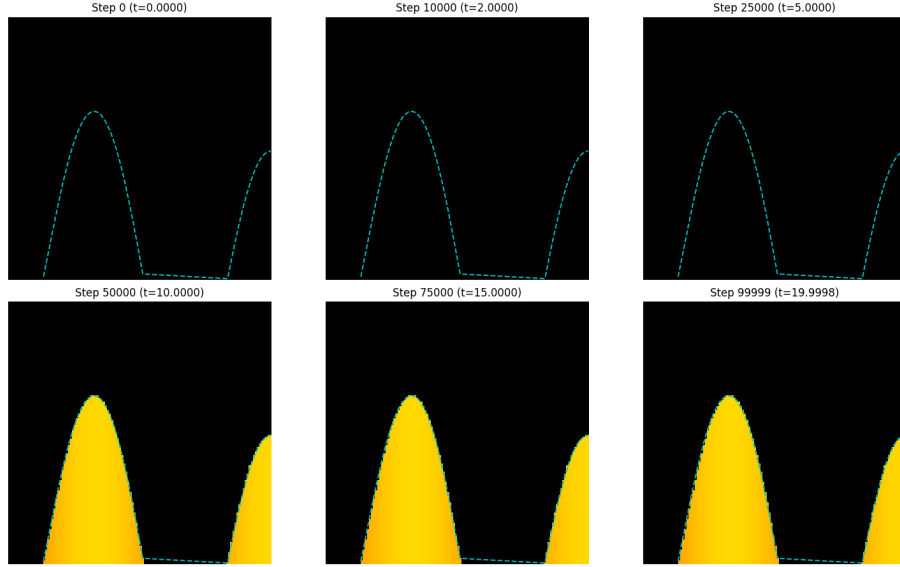


Figure 3: Time Series Plot of Turing Pattern Formation with Planar Threshold

Conclusion

The aim of this work was to examine an assumption made by Sekimura et al. [2]: that the sensitivity of scale cells to morphogen concentrations may vary across the butterfly wing. When constructing or analyzing a model, it is essential to understand the behavior of its individual components before interpreting the combined dynamics. Although the model presented here lacks full biological realism, isolating and studying this single mechanism provides insight into its potential role within a more complete framework. A realistic butterfly wing model ultimately requires the interplay of multiple factors, such as domain geometry and biologically motivated boundary conditions, and the present work serves as a step toward understanding how one such factor contributes to that overall synergy.

References

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Sonnet 4.5 and GPT-4o LLMs were used to generate simulation code.